

3(b)(iii)	See figure above.
4(a)	1. Most alpha-particles passed through undeflected implying that most of the gold foil is empty spaces and thus the nucleus is very small. 2. Some alpha-particles were deflected backwards, that is, 180° but the number is very small suggesting that the size of the nucleus is very small as the possibility of hitting it is very low.

4(b)(i)	$\frac{1}{2}mv^2 = 4.8 \text{ MeV}$ $\frac{1}{2}(4 \times 1.66 \times 10^{-27})v^2 = (4.8 \times 10^{-8}) \times (1.6 \times 10^{-19})$ $v = 1.57 \times 10^7 \text{ m s}^{-1}$
4(b)(ii)	The electric potential V at a point is defined as the work done per unit positive charge in bringing the charge from infinity to that point. A major error was the omission of the charge being positive.
4(b)(iii)	From conservation of energy, Loss in K.E. = gain in E.P.E 4.8 MeV = gain in E.P.E $(4.8 \times 10^{-8}) \times (1.6 \times 10^{-19}) = \frac{(79e)(2e)}{4\pi\epsilon_0 d}$ $d = 4.7 \times 10^{-14} \text{ m}$